



RUTGERS
BIOMEDICAL AND
HEALTH SCIENCES

Methods and applications for multi-study learning with genomics data

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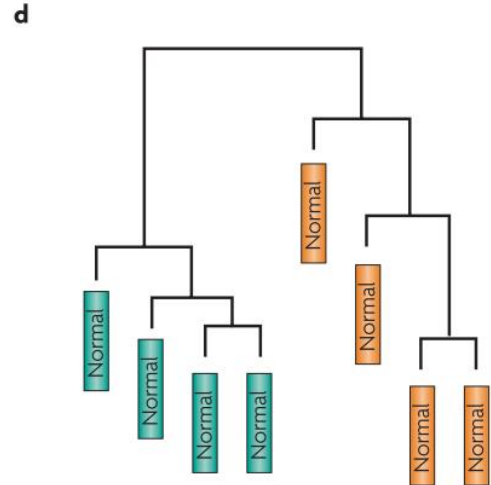
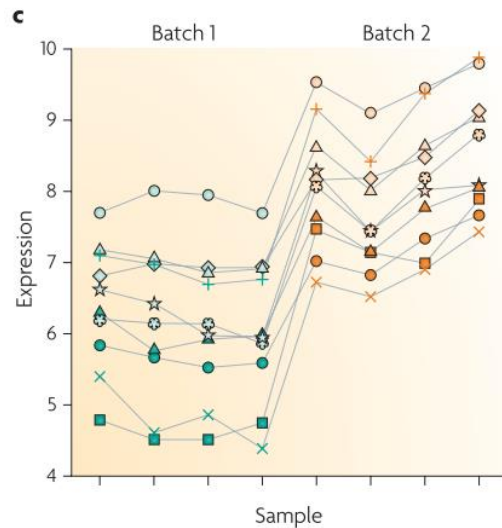
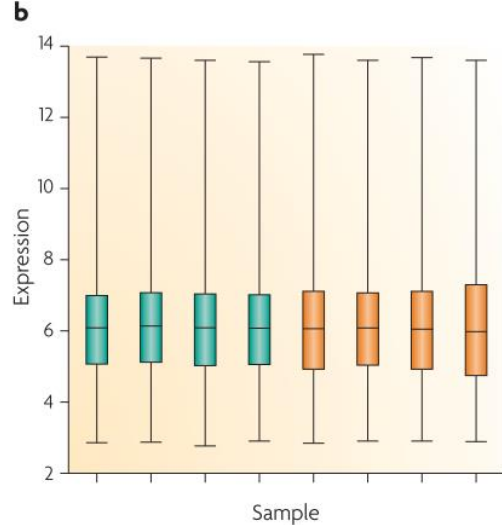
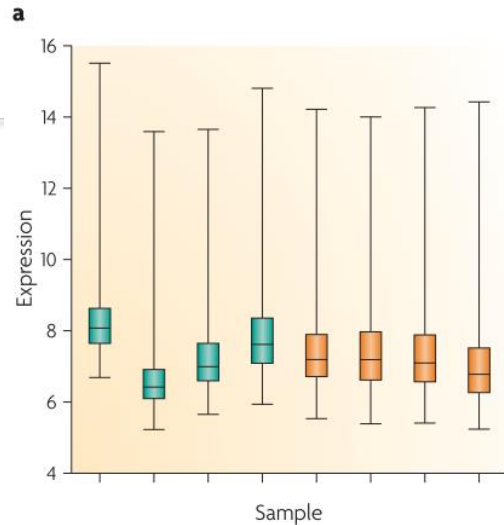


Batch Effects in Genomic Data

- ***Batch Effect:*** Non-biological variation due to differences in batches of data that confound the relationships between covariates of interest.
- Caused by differences:
 - Data profiling/generation platform
 - Lab protocol or experimenter
 - Time of day or hybridization
 - Atmospheric ozone level (Rhodes et al. 2004)

Batch Effect Example: bladder cancer

- Batch effect cannot be fully corrected using standard normalization methods
 - Overall adjustment does not remove all technical variation
 - Genes or cell types impacted differently by batch
 - Coverage differences
 - Library composition batch effect



Combining Batches (ComBat) of Genomic Data

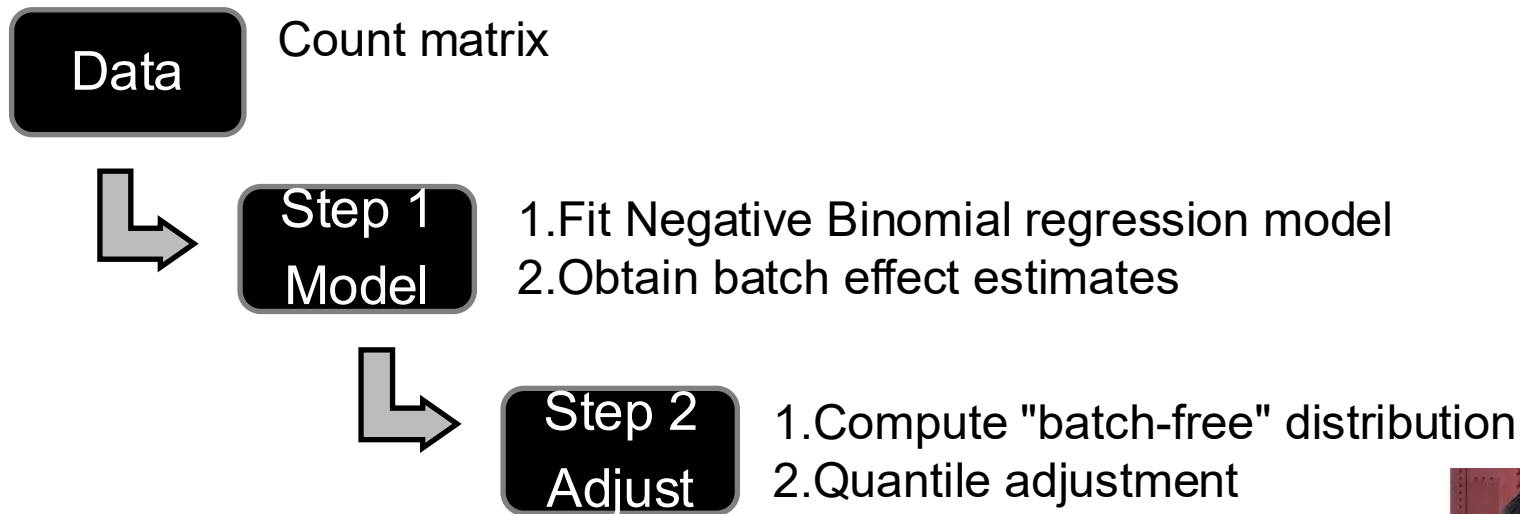
Consider the following model:

$$Y_{ijg} = \alpha_g + X\beta_g + \gamma_{ig} + \delta_{ig}\epsilon_{ijg}$$

where:

- α_g is the overall gene expression
- X is a design matrix
- β_g contains the regression coefficients
- The error terms $\epsilon_{ijg} \sim N(0, \sigma_g^2)$
- γ_{ig} and δ_{ig} are additive and multiplicative batch effects

ComBat-Seq algorithm



Negative Binomial regression

Gene-wise model: for a certain gene g , count in sample j from bat $y_{gij} \sim NB(\mu_{gij}, \phi_{gi})$

$$\log \mu_{gij} = \alpha_g + X_j \beta_g + \gamma_{gi} + \log N_j \quad \text{Var}(y_{gij}) = \mu_{gij} + \phi_{gi} \mu_{gij}^2$$

Decompose scaled
counts into 3
components

$$\left\{ \begin{array}{ll} \alpha_g & \text{Average level for gene } g \text{ (in "negative" samples)} \\ X_j \beta_g & \text{Biological condition of sample } j \\ \gamma_{gi} & \text{Mean batch effect} \end{array} \right.$$

$$\phi_{gi} \quad \text{Variance batch effect}$$



Yuqing Zhang

ComBat-Seq algorithm: Adjust

Adjust the data

Original count matrix

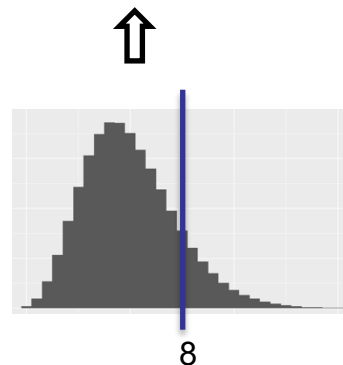
	S 1	S 2
G 1	14	12
G 2	0	0
G 3	112	11
...		

Adjusted count matrix

	S 1	S 2
G 1	9	8
G 2	0	0
G 3	60	47
...		

Empirical
distribution of
original counts:

$$y_{gij} \sim NB(\hat{\mu}_{gij}, \hat{\phi}_{gi})$$



Batch-free
distribution for
adjusted counts:

$$y_{gj}^* \sim NB(\mu_{gj}^*, \phi_g^*)$$



Yuqing Zhang

Research | [Open Access](#) | [Published: 16 January 2020](#)

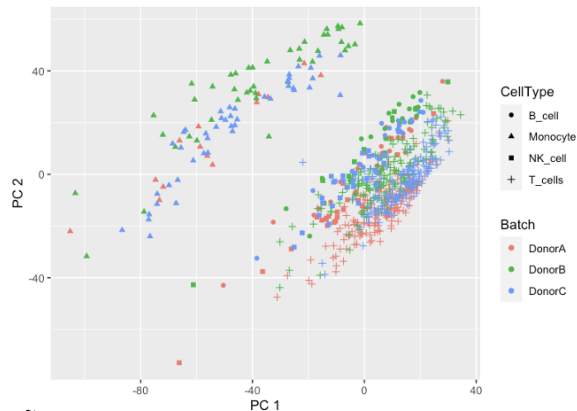
A benchmark of batch-effect correction methods for single-cell RNA sequencing data

[Hoa Thi Nhu Tran](#), [Kok Siong Ang](#), [Marion Chevrier](#), [Xiaomeng Zhang](#), [Nicole Yee Shin Lee](#), [Michelle Goh](#) & [Jinmiao Chen](#) 

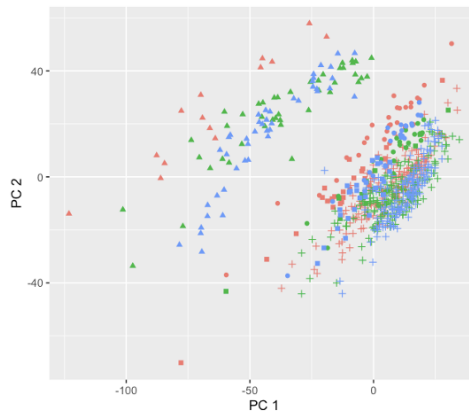
[Genome Biology](#) **21**, Article number: 12 (2020) | [Cite this article](#)

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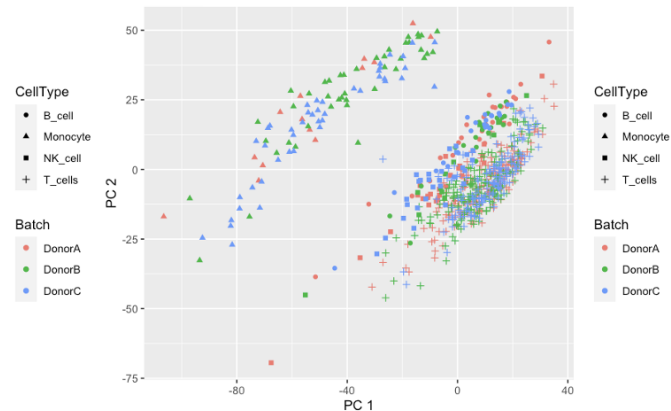
(A) Unadjusted PCA Plot



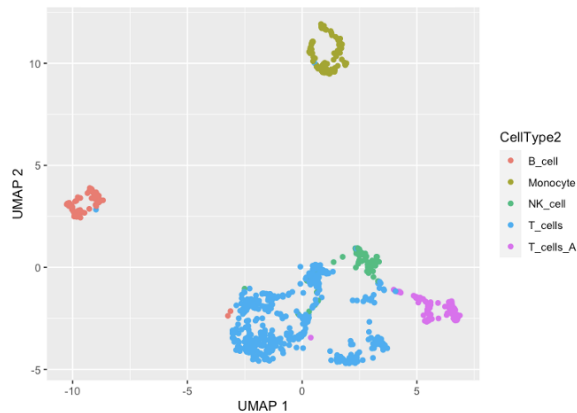
ComBat PCA Plot



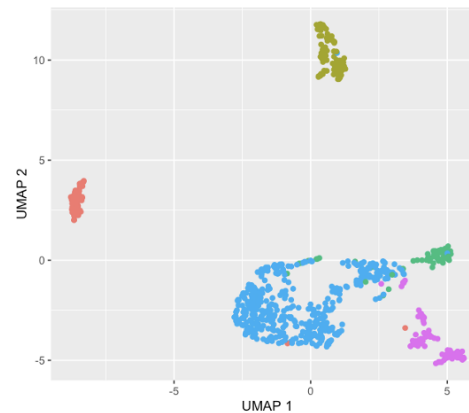
ComBat + Cell Type PCA Plot



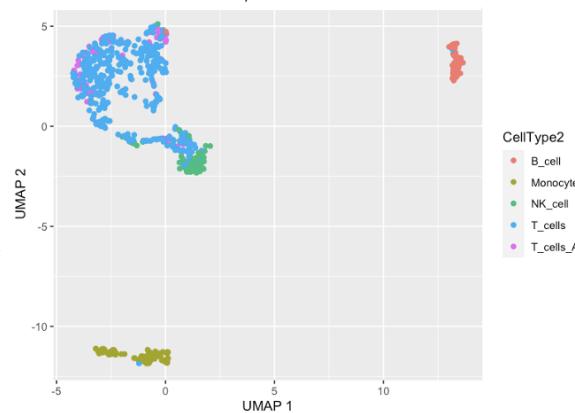
(B) Unadjusted UMAP Plot



ComBat + Cell Type UMAP Plot



Harmony UMAP Plot



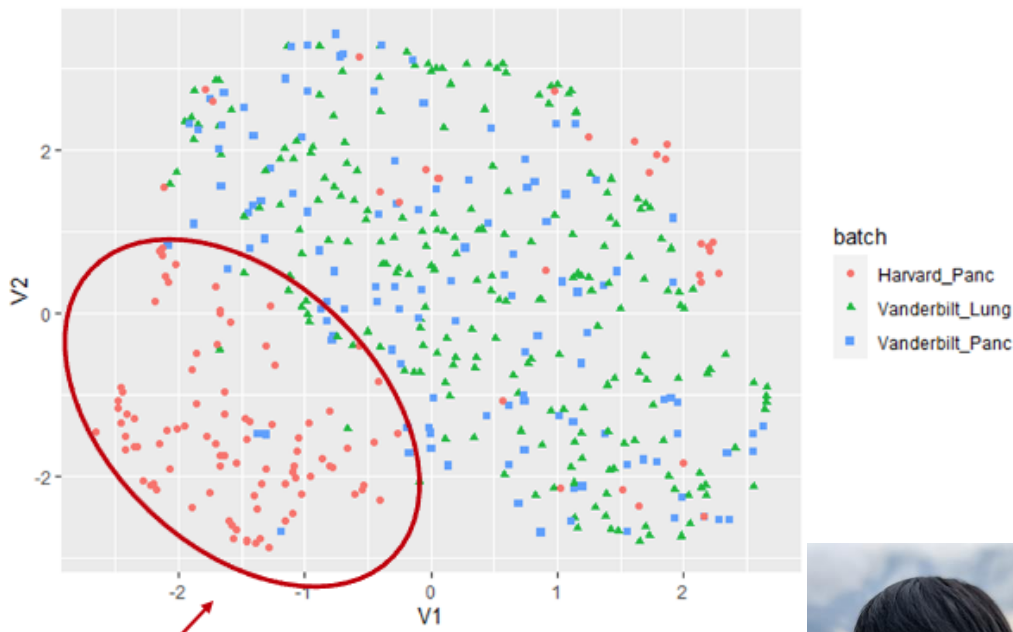
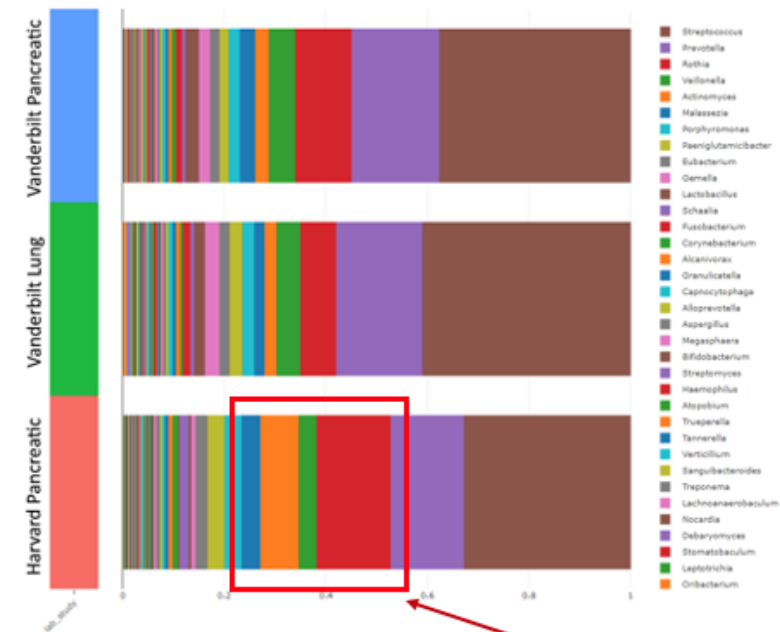
Raw Data Adjustment: Spike in Accuracy

Method	Sensitivity	Specificity
Unadjusted data	0.319	0.869
ComBat-Seq	0.782	0.876
ComBat-Cell-Seq	0.841	0.849
ScMerge	0.718	0.782

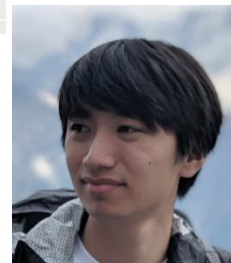
GEE Models for Microbiome Analysis

Relative Abundance Bar Plot

UMAP Plot



Potential Batch Effect



Howard Fan

GEE Models for Microbiome Analysis

- Over-dispersed, zero-inflated, high dimensional, compositional data
- For example, model for Batch effects:

$$Y_{ijg} \sim NB(\mu_{ijg}, \phi_{ig})$$

$$\log \mu_{ijg} = \alpha_g + X_j \beta_g + \gamma_{ig} + \log N_j$$

$$\text{var}(Y_{ijg}) = V_{ij}^{1/2} R_m V_{ij}^{1/2}$$

$$V_{ij} = \text{diag}_m(\mu_{ijm} + \phi_{im} \mu_{ijm}^2)$$

R_m describes compositional nature

- Methods and tools for longitudinal microbiome data



Yuqing Zhang



Brie Odom



Howard Fan



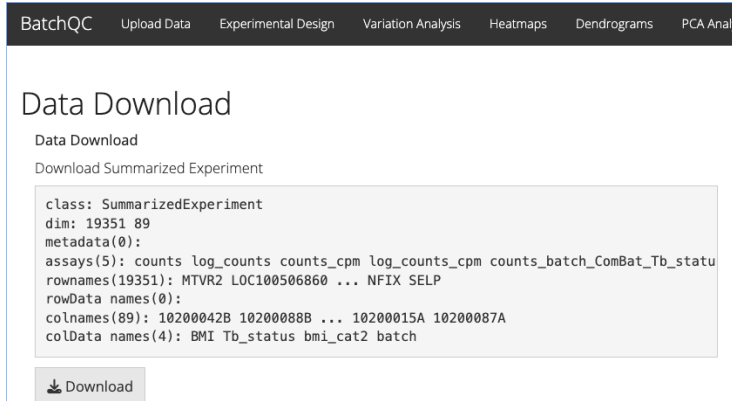
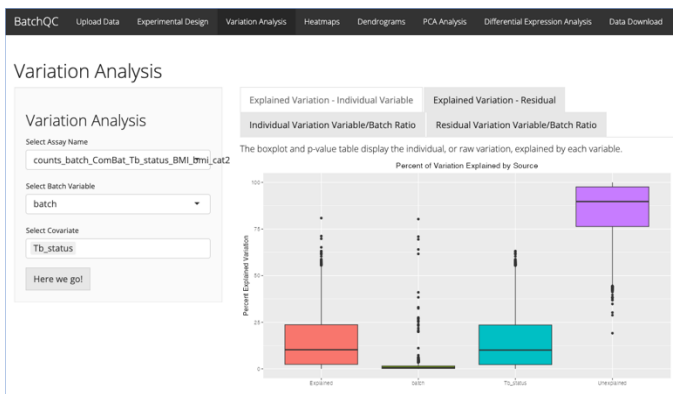
Lucas Schiffer

BatchQC: Evaluating Batch Effects

- Test for Negative Binomial Assumption
- Apply multiple batch correction methods (including ComBat-Seq)
- Evaluate need for batch
- Interactive Visualizations
- Compare original and batch corrected data side by side
- Export corrected data
- Shiny App or R functions



Jessica McClintock

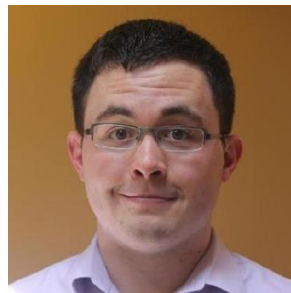


Solaiappan Manimaran

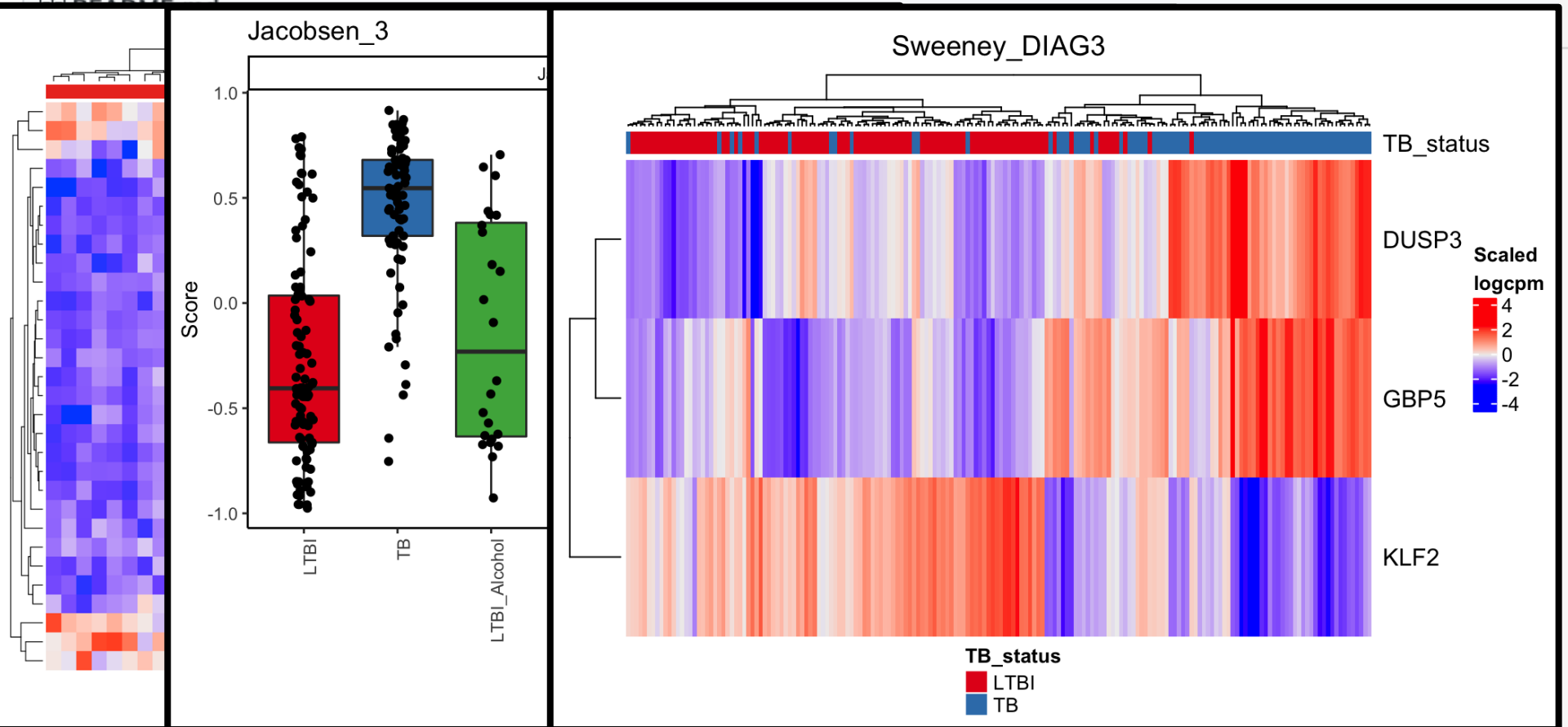
- R Package for profiling 70+ published signatures
 - Signatures of disease, response, and progression
- Scoring with original models (**new!**) and gene set enrichment methods:
 - GSVA, ssGSEA, PLAGE, singscore, zscore, and ASSIGN
- Visualization tools and functions:
 - Annotated heatmaps, boxplots, ROC, p-value and AUC tables
- **NEW!** TBSignatureProfiler graphical interface



Brie Odom



David Jenkins



RUTGERS The curatedTBData Project

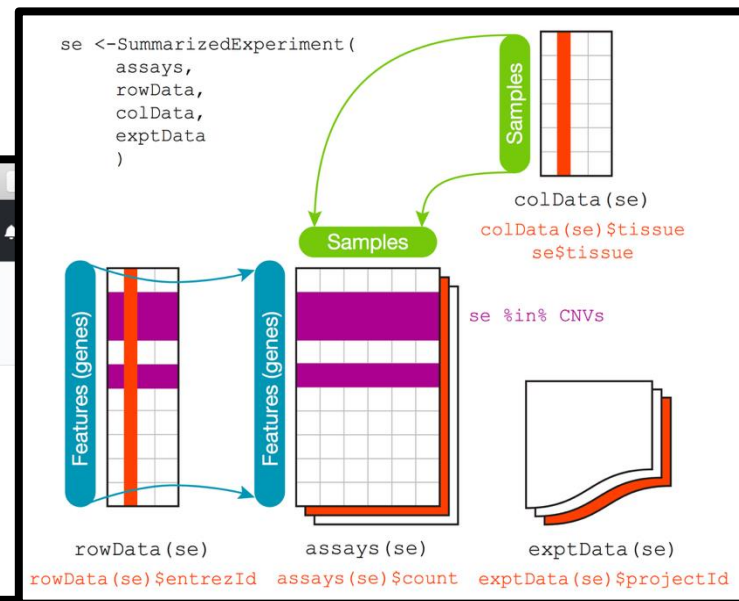
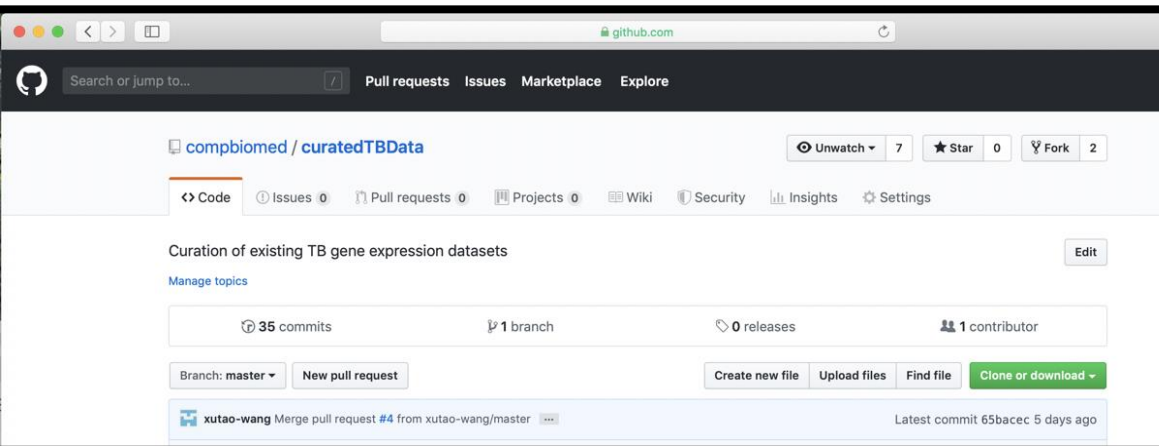
- Curate expression data: currently 42+ TB experiments—more than 4,000 samples!
- Multiple platforms (array, sequencing, etc)
- Using SummarizedExperiment Objects in R
- Resource for biomarker validation/development!



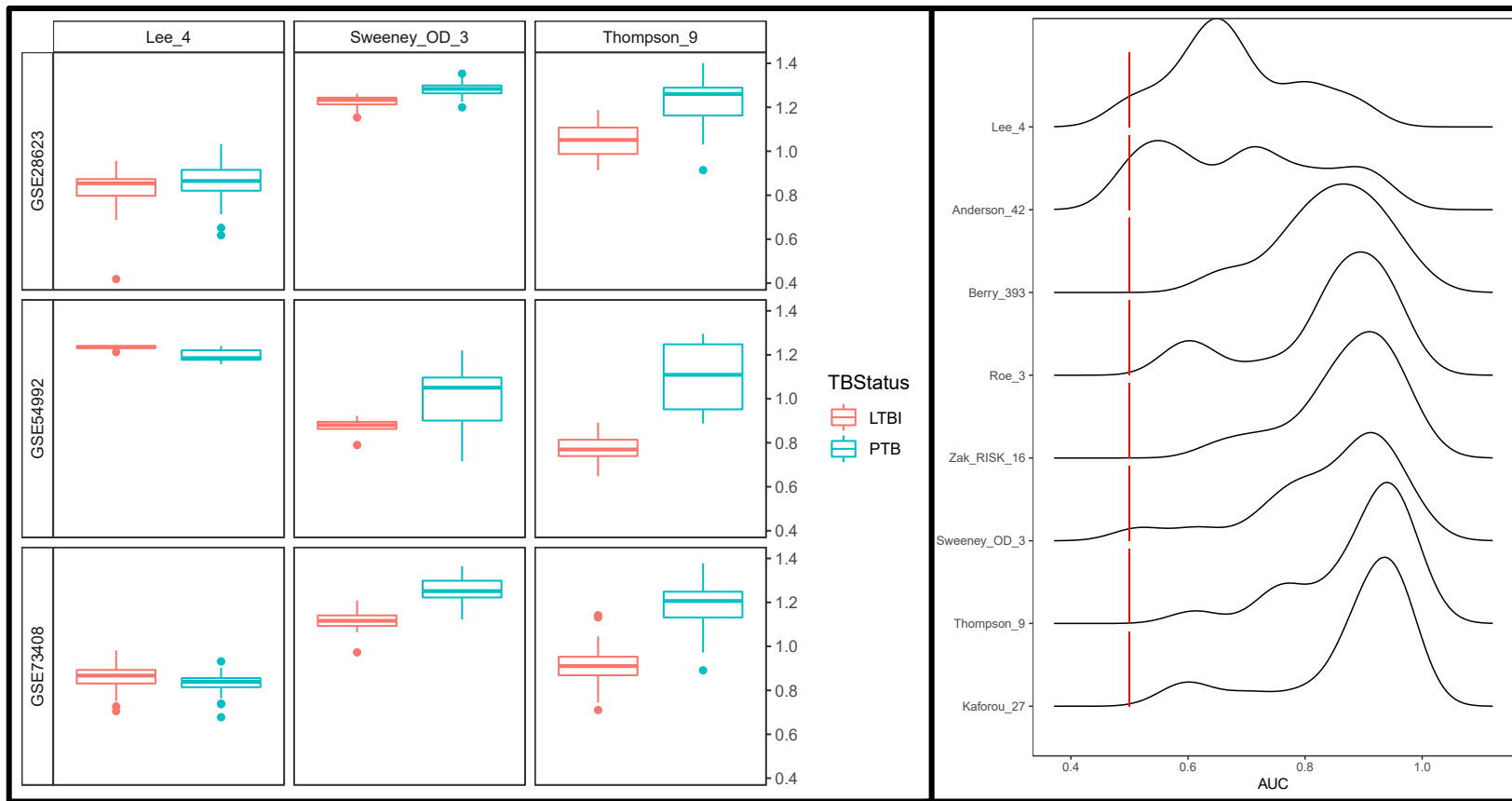
Dr. Prasad Patil



Xutao Wang

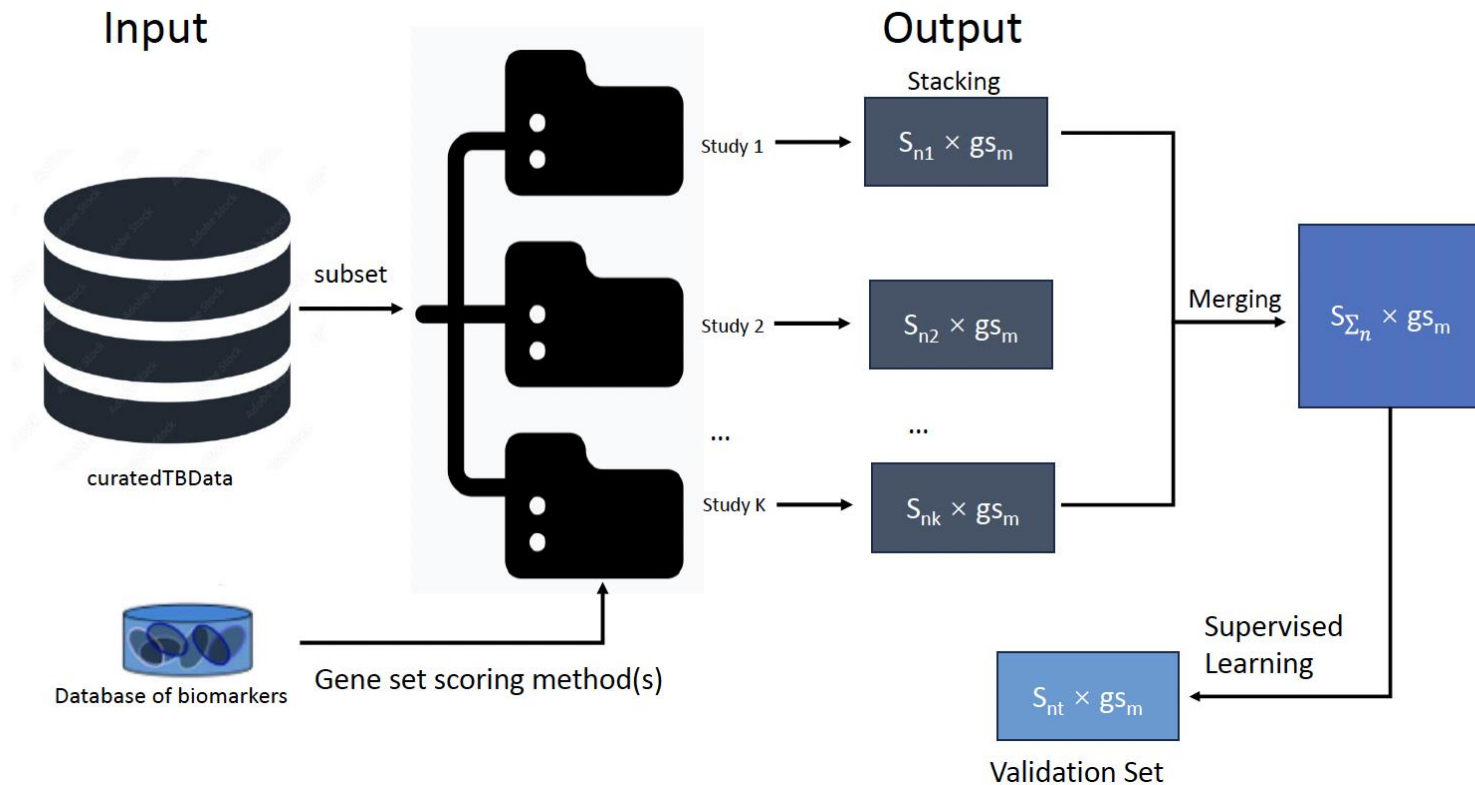


<https://github.com/compbiomed/curatedTBData>



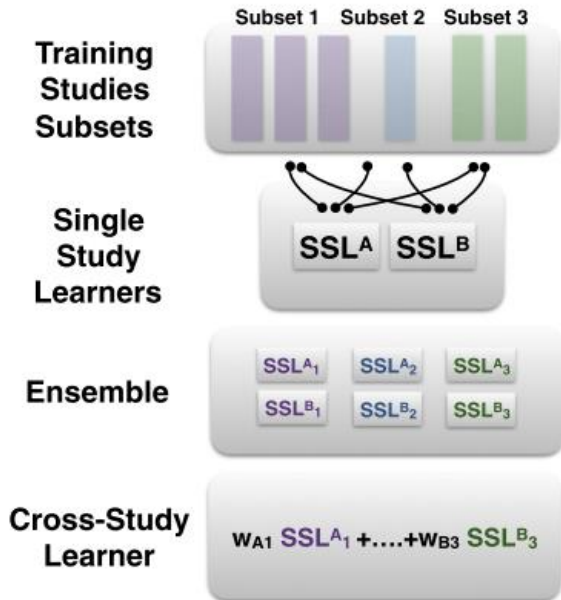


Goal: Improve the predictive ability of existing biomarkers using cross study learning.



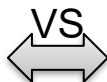
Solution when multiple studies available: ensemble learning

- Ensemble learning:
 - Established machine learning approach
 - Combined multiple models to improve prediction
- Cross-study learners (CSL)
 - Generalizable performance in independent test set
 - Experimental and theoretical insights (regression)



Cross-study learners from Patil and Parmigiani, "Training replicable predictors in multiple studies", PNAS 2018

Intuition

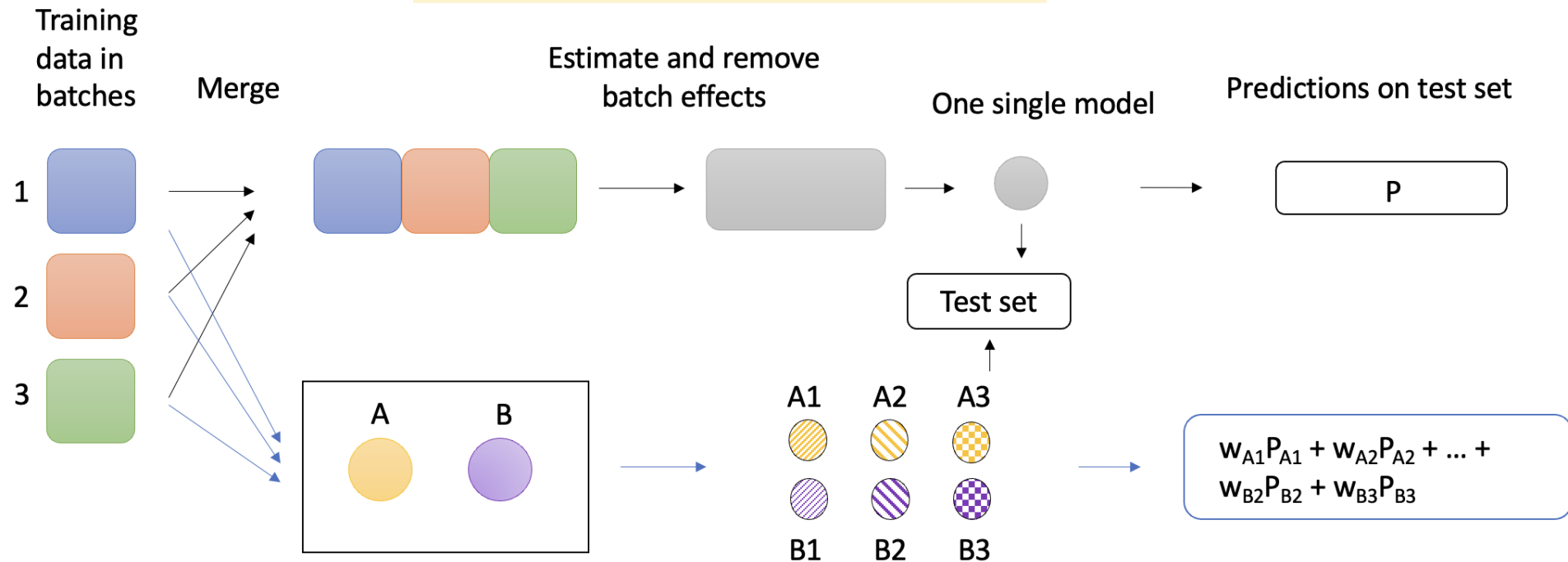


- Integrates data
- Remove undesired variation from as many of the genomic features as feasible, then use the "cleaned" data in training

- Integrates models
- Weights reward prediction functions that, while trained in one batch, continue to predict well in other batches
- likely to avoid using features affected by batch effect

Should we Merge or Ensemble?

Merge followed by batch correction (Merging)



Bioinformatics

Issues

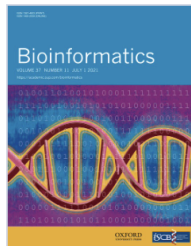
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Volume 37, Issue 11
June 2021

Article Contents

Abstract

1 Introduction

JOURNAL ARTICLE

Robustifying genomic classifiers to batch effects via ensemble learning ^{FREE}

Yuqing Zhang, Prasad Patil, W. Evan Johnson, Giovanni Parmigiani ✉

Bioinformatics, Volume 37, Issue 11, June 2021, Pages 1521–1527,

<https://doi.org/10.1093/bioinformatics/btaa986>

Published: 27 November 2020

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Abstract

Motivation



Yuqing Zhang

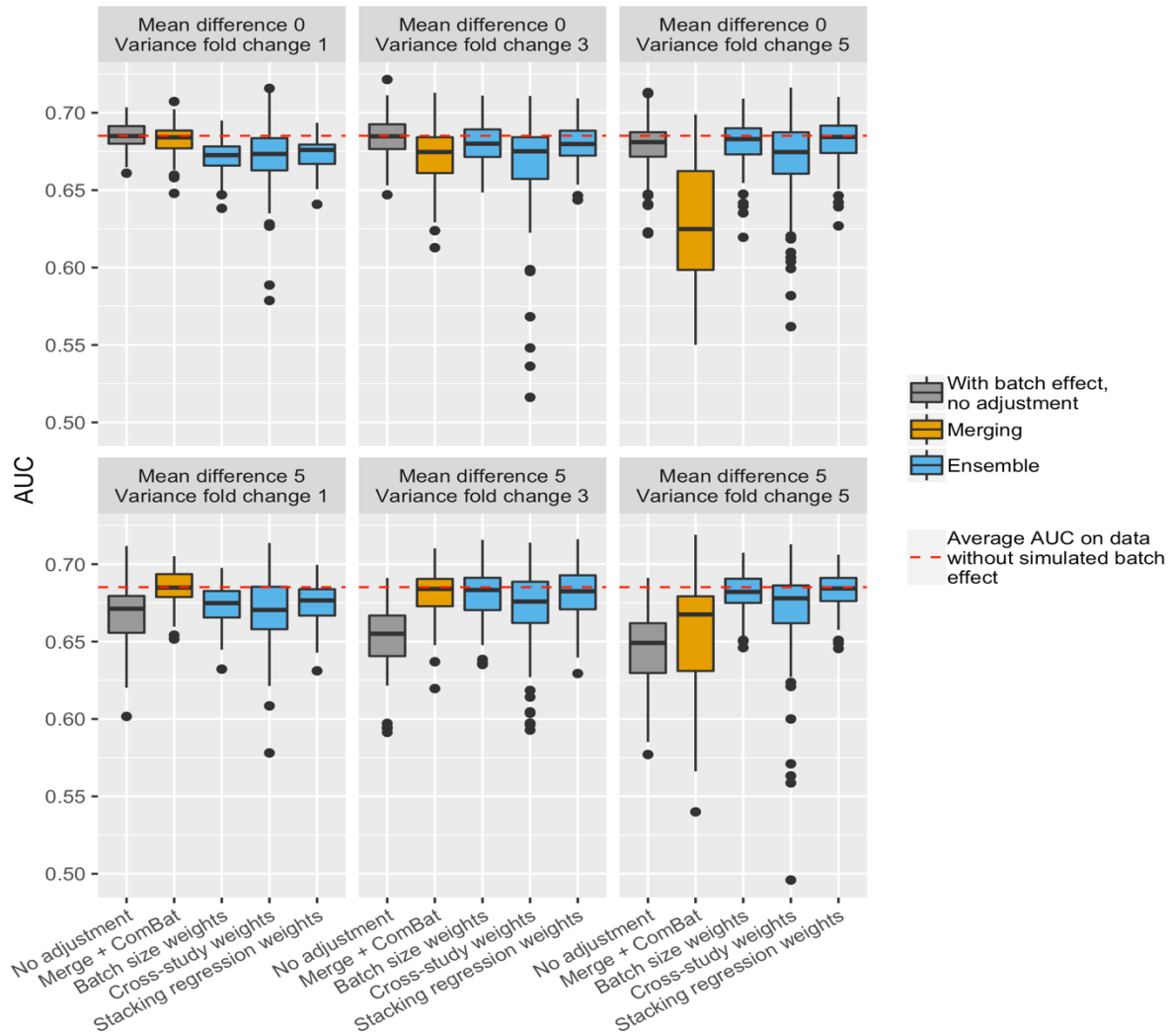


Prasad Patil



Giovanni Parmigiani

- Example results using random forest (RF) as learner
- Transition point: merging out-performs ensemble at low level of batch effect; ensemble is better at high batch differences
- Consistent with established cross-study results



Traditional models:

Training and testing datasets are distributed identically. Violated by:

- covariate shift
- distribution gap

Multi-study learning

Common for multiple datasets with shared covariates and outcomes:

- Inefficient to rely on one data source for training

Multi-source domain adaptation:

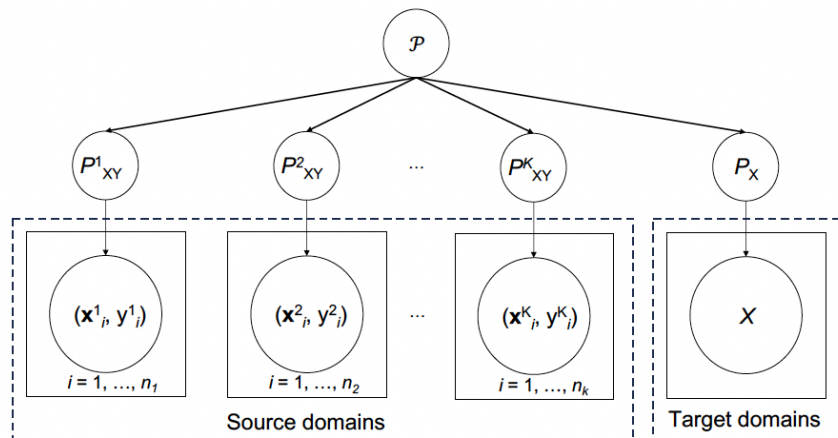
- Merge all training studies
- Keep each training dataset separately



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- Denote $\mathcal{X}_S^j = \{(\mathbf{x}_i^j, y_i^j)\}_{i=1}^{n_j}$ as the j -th source/training domains, and $\mathcal{X}_T$ as target/testing domain with inaccessible label.
- Learn a robust and generalizable prediction function f from the K source domains s.t. $\min_f E_{(\mathbf{x}, y) \in \mathcal{X}_T} [\ell(f(\mathbf{x}), y)]$.

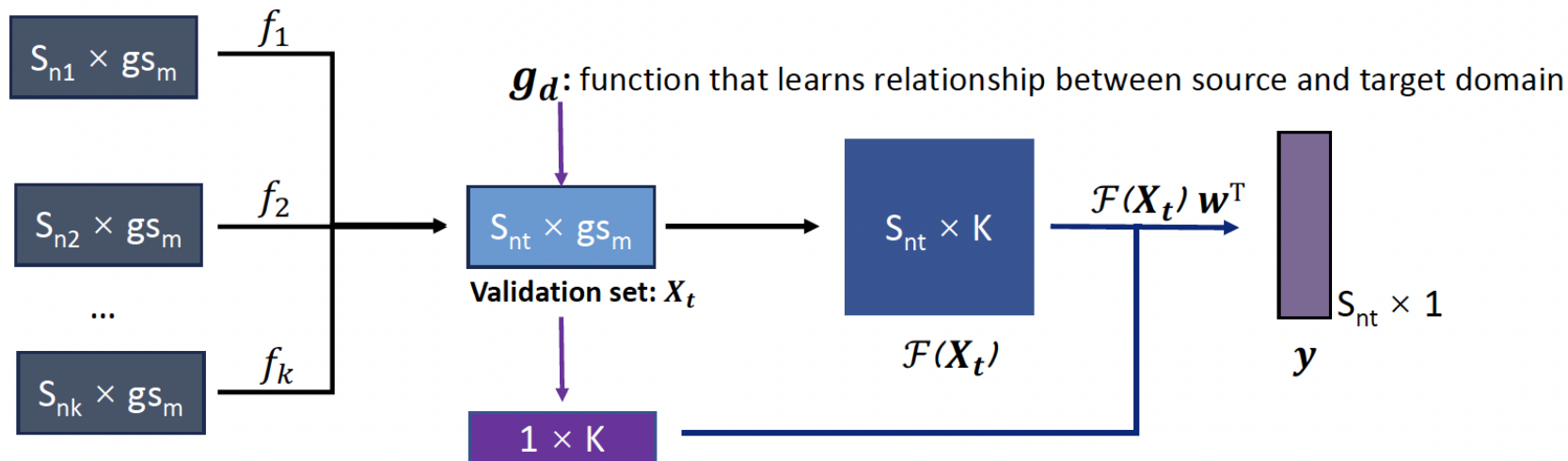


(Wang et al., *IEEE Trans. Knowl. Data Eng.*, 2022)

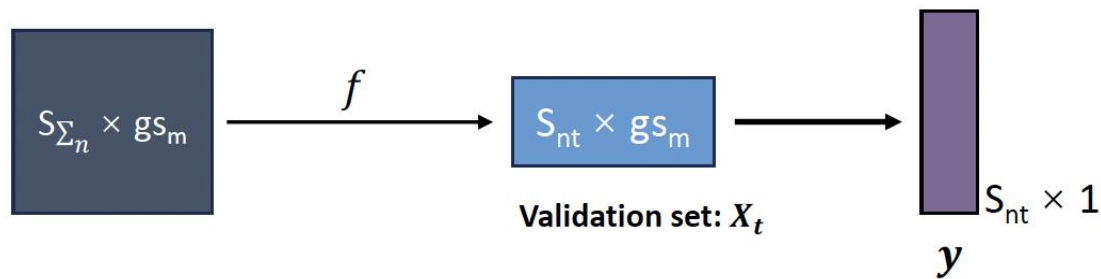
Integrating source-based classifiers:

$$y_i = \mathcal{F}(\mathbf{x}_i) = \sum_{j=1}^K w_j f_j(\mathbf{x}_i), \quad \sum_j w_j = 1$$

Cross study learning:



Merging:



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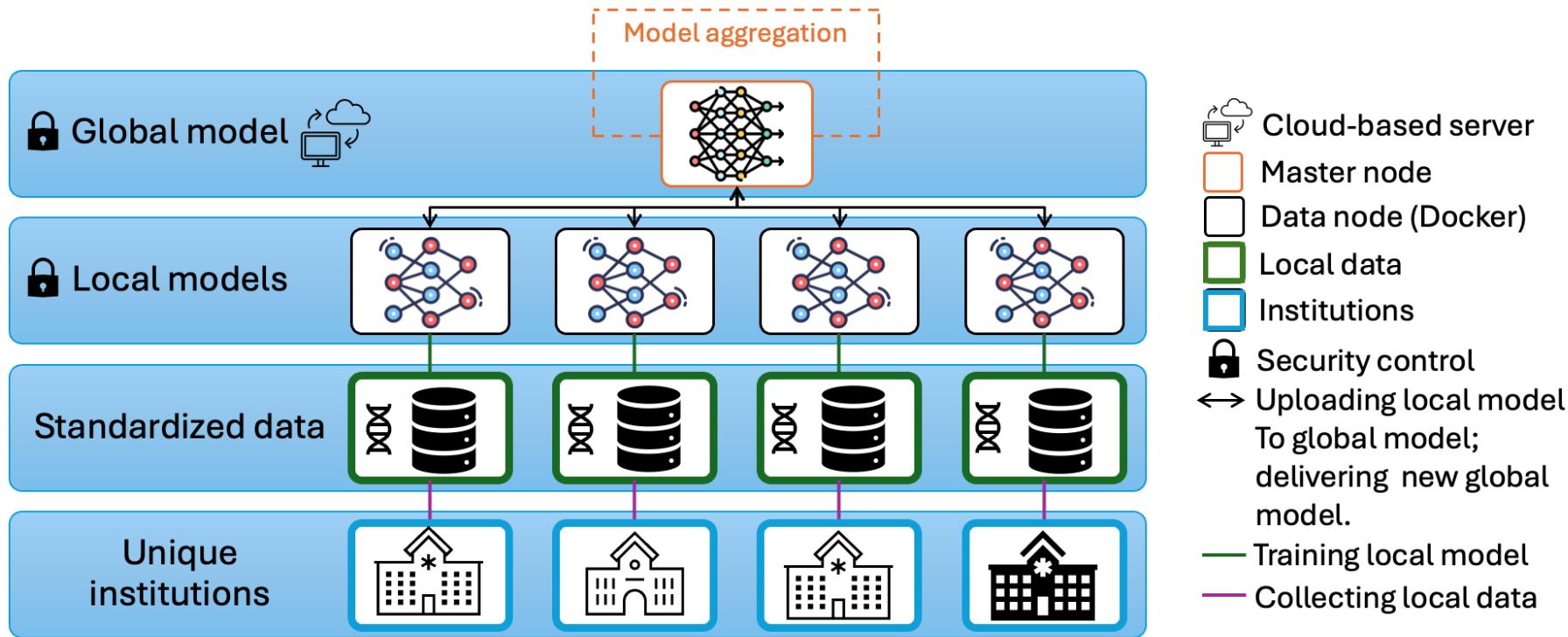
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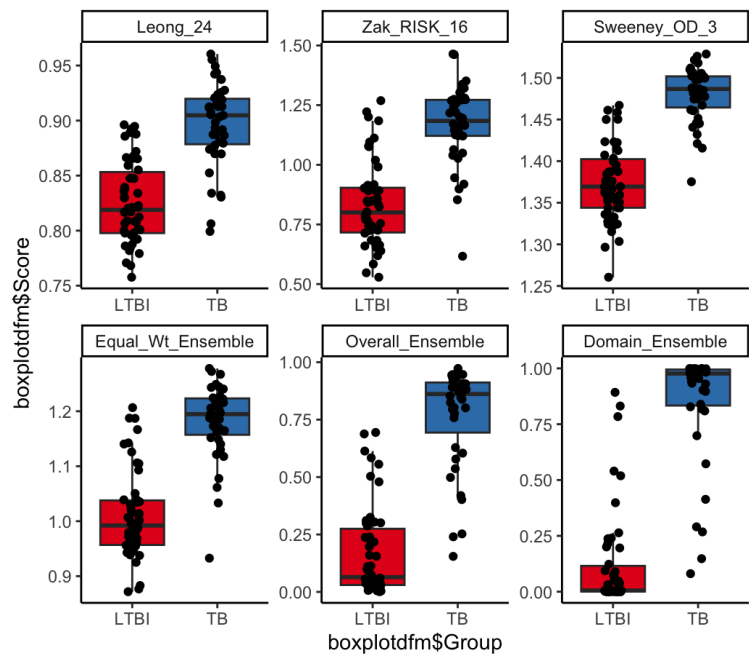
RUTGERS Adaptive Learning Improves Prediction

Comparison	Method	Accuracy	Sensitivity	Specificity
PTB vs. HCs	Adaptive weight	0.84 (0.80-0.89)	0.84 (0.76-0.92)	0.89 (0.83-0.93)
	Equal weight	0.84 (0.80-0.89)	0.84 (0.76-0.92)	0.89 (0.83-0.93)
	Sample-size weight	0.84 (0.80-0.90)	0.81 (0.72-0.90)	0.92 (0.89-0.96)
	Stacking	0.85 (0.80-0.91)	0.85 (0.78-0.92)	0.89 (0.84-0.95)
PTB vs. Others	Adaptive weight	0.81 (0.77-0.84)	0.77 (0.71-0.82)	0.85 (0.81-0.90)
	Equal weight	0.80 (0.76-0.84)	0.77 (0.72-0.82)	0.84 (0.79-0.89)
	Sample-size weight	0.75 (0.71-0.80)	0.56 (0.48-0.64)	0.92 (0.88-0.95)
	Stacking	0.76 (0.72-0.79)	0.56 (0.50-0.63)	0.93 (0.90-0.96)
PTB vs. OD	Adaptive weight	0.69 (0.65-0.74)	0.66 (0.57-0.76)	0.76 (0.71-0.81)
	Equal weight	0.67 (0.63-0.72)	0.65 (0.58-0.72)	0.75 (0.68-0.80)
	Sample-size weight	0.67 (0.62-0.72)	0.60 (0.51-0.69)	0.79 (0.73-0.83)
	Stacking	0.57 (0.48-0.67)	0.30 (0.13-0.53)	0.90 (0.83-0.96)

RUTGERS Federated Learning: Architecture

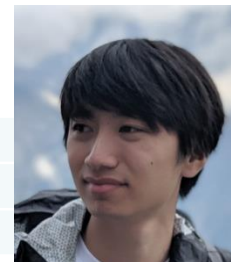
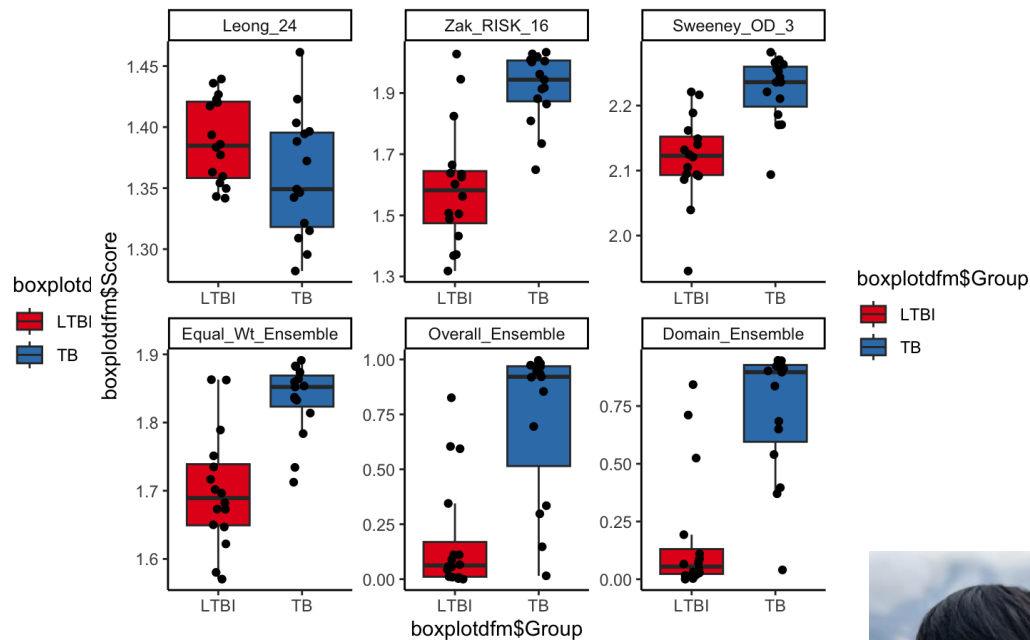


Indian Dataset



Country	Equal Weights	Ensemble Weights	Domain Weights
India	0.93	0.96	0.97
Africa	0.88	0.90	0.91

Ugandan Dataset



Howard Fan



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